

WASP-10b

R-0822

Benchmark run · workflow W8-benchmark-deep · record deep-bench_WASP-10_171027 · created 2026-07-28T15:15:55+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2458053.6685 +/- 0.0016 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1643 +/- 0.0084
Transit depth (Rp/Rs)^2	2.7 +/- 0.28 [%]
Orbital Inclination (inc)	88.16 +/- 0.67
Airmass coefficient 1 (a1)	1.0127 +/- 0.0087
Airmass coefficient 2 (a2)	-0.013 +/- 0.04
Scatter in the residuals of the lightcurve fit is	0.86 %
Best Comparison Star	#3 - [352, 131]
Optimal Aperture	3.7
Optimal Annulus	12.34
Transit Duration (day)	0.0903 +/- 0.004

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1643 ± 0.0084 vs 0.1592 (deviation 0.61σ)
Duration fit vs published	0.0903 d vs 0.0946 d (deviation 1.07σ)
Mid-time O–C	0.7 min
Depth SNR	19.6
Residual scatter	0.86 %

Light curve

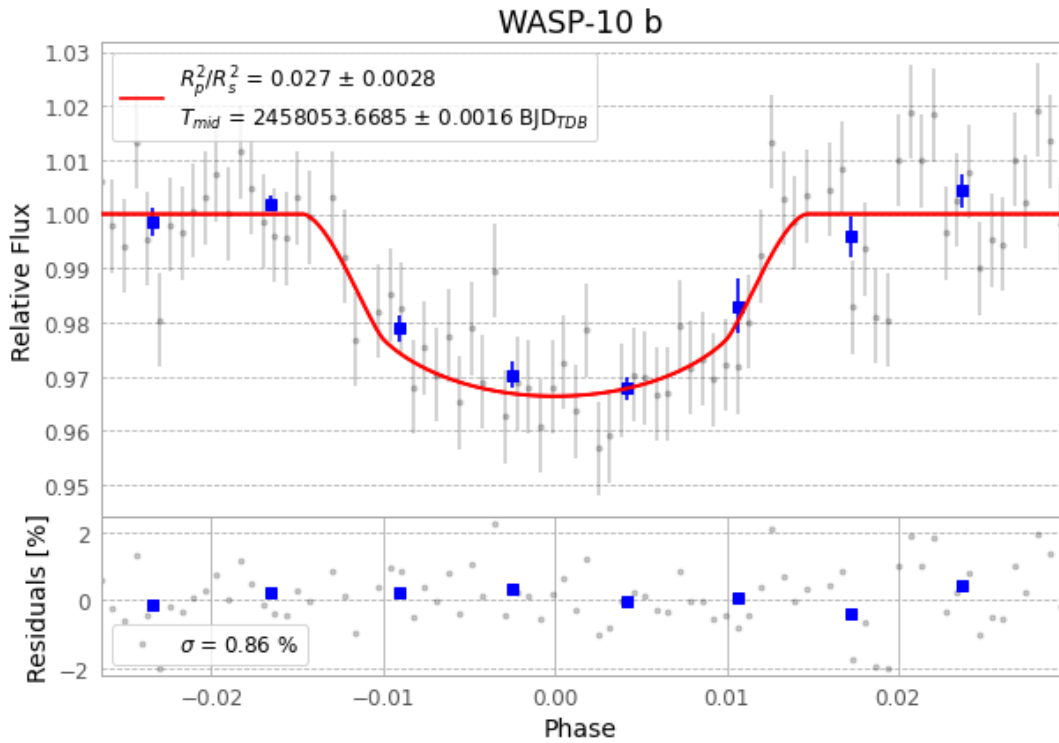


Figure 1 — FinalLightCurve_WASP-10 b_2017-10-26.png (SHA-256 3c51ac85cb32082c...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [356, 231], 3 comparison star(s). Exact configuration: parameters.inits.json in record.json (SHA-256 0555ff322df31989...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

84 FITS frames (55 MB), WASP-10171027015712.FITS → WASP-10171027060612.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-10-171027.log (43ac39e6bd2f...), FinalLightCurve_WASP-10 b_2017-10-26.png (3c51ac85cb32...), FinalLightCurve_WASP-10 b_2017-10-26.csv (02bdb64b70c2...)

Compute resources

Wall time 190.4 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 15:15Z.